

Human posture recognition using human skeleton provided by Kinect

Thi-Lan Le

International Research Institute MICA
HUST - CNRS/UMI-2954 -
GRENOBLE INP
HANOI UNIVERSITY of SCIENCE
and TECHNOLOGY
VIET NAM

Minh-Quoc Nguyen

International Research Institute MICA
HUST - CNRS/UMI-2954 -
GRENOBLE INP
HANOI UNIVERSITY of SCIENCE
and TECHNOLOGY
VIET NAM

Thi-Thanh-Mai Nguyen

International Research Institute MICA
HUST - CNRS/UMI-2954 -
GRENOBLE INP
HANOI UNIVERSITY of SCIENCE
and TECHNOLOGY
VIET NAM

Abstract— Human posture recognition is an attractive and challenging topic in computer vision because of its wide range of application. The coming of low cost device Kinect with its SDK gives us a possibility to resolve with ease some difficult problems encountered when working with conventional cameras. In this paper, we explore the capacity of using skeleton information provided by Kinect for human posture recognition in a context of a health monitoring framework. We conduct 7 different experiments with 4 types of features extracted from human skeleton. The obtained results show that this device can detect with high accuracy four interested postures (lying, sitting, standing, bending).

Index Terms—Human posture recognition, human skeleton, Kinect, SVM

I. INTRODUCTION

Human posture recognition (HPR) is an attractive topic in computer vision because of its wide range of application. HPR can be viewed as a sub-field of gesture recognition since a posture is a “static gesture”. In practice, posture recognition is usually at the crossroad between people detection and gesture recognition. Sometimes we are only interested in the posture at a given time which can be performed by a people detector [1]. In other cases, posture detection can be sometimes considered as a first step for gesture recognition, for instance, by associating postures to states of a Finite State Machine (FSM) [2]. The challenges of posture recognition are seamlessly the same as gesture recognition except that the temporal aspect is not accounted.

In our work, we are interested in detecting key human postures for abnormal event modeling and recognition in a health monitoring system. In the literature, there are a number of works that have been proposed for HPR using conventional cameras. Recently, Kinect device has been released allows to capture not only color information as conventional camera but also depth and motion information. This device becomes quickly popular because of its low cost as well as its free SDK. In this paper, we explore the possibility of using Kinect device for HPR problem.

II. RELATED WORKS

In the literature, there are a number of works that have been proposed for human posture recognition. However, most of the works use color information captured from traditional cameras.

Cohen and Li [3] classified body posture with SVM technique from a 3D visual hull constructed from a set of silhouette input data. The system returns the recognized human postures in the form of thumbnail images. Cheng Mo et al. [4] proposed a human behavior analysis system which recognizes human postures as walking, bending down and sitting. A multiclass SVM is used to classify human posture using human skeleton, angles of six sticks in the skeleton and object motion vectors. Zhao and Liu [5] used a centroid radii model as shape descriptor to represent human posture in each frame. Non-linear SVM decision tree is used to recognize human postures such as standing, lying, bending, and sitting. In [6], Chella et al. proposed a system for simultaneous people tracking and posture recognition in cluttered environments in the context of human-robot interaction. As soon as the tracking algorithm individuates a person to track, the system also estimates its posture. The recognition adopts a modified eigen-space technique in order to classify seven different postures: standing, pointing left, pointing right, stop (pointing both left/right), left arm raised, right arm raised, both arms raised. In [7], a combination of MPEG-7 contour-based shape descriptor and projection histogram was used to recognize the main postures and the view of a human based on the binary object mask obtained by the segmentation process. The recognition is treated as a typical pattern recognition task and is carried out through a hierarchy of classifiers. The system can recognize four main postures: standing, sitting, bending and laying with four views: front, back, left, right.

The above mentioned human posture recognition works are dedicated to color camera. In these works, in order to recognize human posture, human region in image has to be determined. The main weak point of these works is that they are sensitive to the change of clothing and lighting condition. Unlike these works, the work proposed in this paper aims at detecting human posture from the skeleton (that is defined by using

infrared laser projector), it therefore is invariant to the change of clothing and lighting condition.

III. KINECT BASED HUMAN POSTURE RECOGNITION

Our architecture system is presented in Fig. 1. This system consists of 3 main modules: data acquisition, data processing and feature extraction, human posture recognition. In the data acquisition module, we can capture different types of information: color, depth and skeleton information. The data processing and feature extraction module aims at doing some processing if needed such as data normalization and at computing relevant features for posture representation. It is worth to note that, besides the depth and skeleton information, Kinect provides color image as conventional camera, all features proposed for the conventional camera can be applied for Kinect device. However, in this paper, we focus on analyzing the possibility to use only skeleton information for human posture recognition. The human posture recognition module aims at learning and classifying given posture in one of the predefined classes such as lying, bending.

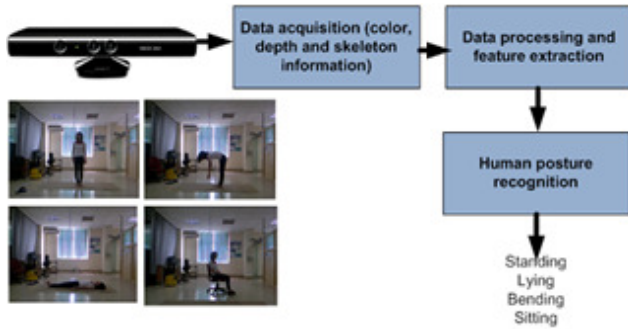


Fig. 1. Main modules in our proposed system

A. Data acquisition

The Kinect device consists of an infrared laser projector combined with a monochrome CMOS sensor which captures video data. This device has a RGB camera and a multi-array microphone. Therefore, the Kinect device offers to us the possibility to capture at the same time the color image and depth image of the observed scene. Fig. 2 shows an image of Kinect device.

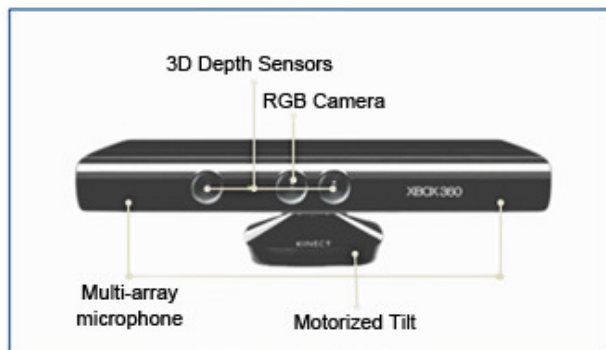


Fig. 2. Kinect device consists of an infrared laser projector combined with a monochrome CMOS sensor, RGB camera and a multi-array microphone

Recently, in the new version of Kinect SDK, a skeletal tracking tool has been provided. This tool aims at collect the joints as points relative to the device itself. The joint information is collected in frames. For each frame, the positions of 20 points are estimated and collected. The 20 joints are shown as Fig. 3.

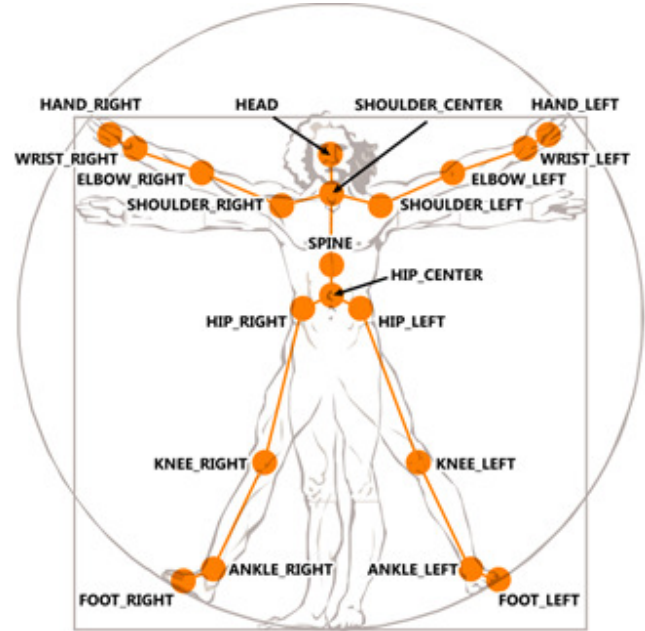


Fig. 3. Human skeleton

For each joint, we have three main information. The first information is the index of the joints. Each joint has a unique index value. The second information is the positions of each joint in x, y, and z coordinates. These three coordinates are expressed in meters. The x, y, and z axes are the body axes of the depth sensor. This is a right-handed coordinate system that places the sensor array at the origin point with the positive z axis extending in the direction in which the sensor array points. The positive y axis extends upward, and the positive x axis extends to the left (with respect to the sensor array). The three coordinates for joint position are presented in Fig. 4.

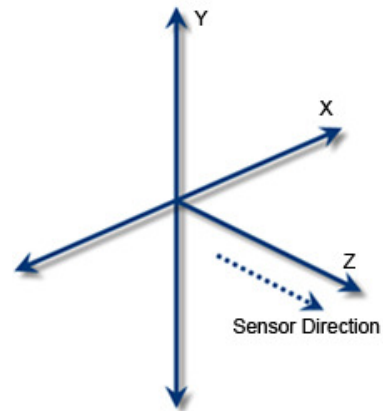


Fig. 4. Three coordinates for joint position

The last information is the status of the joint. If Kinect is able to track the joint, it set the status of this joint 'tracked'. In the case if the joint cannot be tracked, the algorithm tries to infer the joint position from other joints. If possible, then the status of this joint is inferred. Otherwise, the status of the joint is non-tracked.

An example of color, depth image and skeleton captured from Kinect for standing posture is illustrated in Fig. 5.



Fig. 5. Color, depth and skeleton captured from Kinect for standing posture

B. Data processing and Feature extraction

With the skeleton tracked by Kinect, the first feature can be extracted is joint positions. Since, each joint has 3 values of 3 coordinates and a skeleton consist of 20 joints. So, the feature vector has 60 dimensions. When working with predefined postures, we can choose important joints for representing these postures. Besides that feature, other features can be derived from joint position such as joint angles. When working with four postures (sitting, lying, standing, bending), we observe that 10 joints (A, B, C, D, E, F, G, H, O and Q) are the most important joints for representing these postures. From these joints, we can calculate different sets of angles. In the experimentation section, we will analyze the recognition performance using these angles.

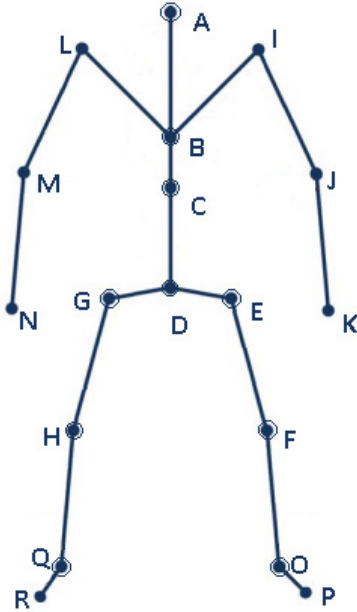


Fig. 6. Important joints defined for 4 main postures representation

C. Human posture recognition

We propose to use SVM (Support Vector Machine) to recognize the human postures from the extracted feature. SVM

was selected for classification in our research due to high accuracy and ability to work with high dimensional data, ability to generate non-linear and well as high dimensional classifier.

Let $\{x_i, y_i\}, i = 1, \dots, l, p, y_i \in \{-1, 1\}, x_i \in R^d$ be the training data with labels y . The support vector machine (SVM) using C-Support Vector Classification (C-SVC) algorithm will find the optimal hyper-plane:

$$f(x) = w^T \Phi(x) + b \quad (1)$$

to separate the training data by solving the following optimization problem:

$$\min \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (2)$$

subject to

$$y_i [w^T \Phi(x_i) + b] \geq 1 - \xi_i \text{ and } \xi_i \geq 0, i = 1, \dots, l \quad (3)$$

The optimization problem (2) will guarantee to maximize the hyper-plane margin while minimize the cost of error. $\xi_i \geq 0, i = 1, \dots, l$ are non-negative slack variables introduced to relax the constraints of separable data problem to the constraint of non-separable data problem. For an error to occur the corresponding ξ_i must exceed unity (3), so ξ_i is an upper bound on the number of training errors. Hence an extra cost $C \sum_i \xi_i$ for errors is added to the objective function (2) where C is a parameter chosen by the user.

The Lagrangian formulation of the primal problem is:

$$\begin{aligned} L_p = & \frac{1}{2} \|w\|^2 + C \sum_i \xi_i \\ & - \sum_i \alpha_i \{y_i (x_i^T w + b) - 1 + \xi_i\} \\ & - \sum_i \mu_i \xi_i \end{aligned} \quad (4)$$

We will need the Karush-Kuhn-Tucker conditions for the primal problem to attain the dual problem:

$$L_D = \sum_i \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \Phi(x_i)^T \Phi(x_j) \quad (5)$$

Subject to:

$$0 \leq \alpha_i \leq C \text{ and } \sum_i \alpha_i y_i = 0 \quad (6)$$

The solution is given by:

$$w = \sum_i^{N_s} \alpha_i y_i x_i \quad (7)$$

Where N_s is the number of support vectors.

Note that data only appear in the training problem (4) and (5) in the form of dot product $\Phi(x_i)^T \Phi(x_j)$ and can be replaced by any kernel K with $K(x_i, x_j) = \Phi(x_i)^T \Phi(x_j)$ is a mapping to map the data to some other (possibly infinite dimensional) Euclidean space. One example is Radial Basis Function (RBF) kernel $K(x_i, x_j) = e^{-\gamma \|x_i - x_j\|^2}$.

In test phase an SVM is used by computing the sign of

$$f(x) = \sum_i^{N_s} \alpha_i y_i \Phi(s_i)^T \Phi(x) + b \quad (8)$$

$$= \sum_i^{N_s} \alpha_i y_i K(s_i, x) + b$$

where the s_i are the support vectors.

IV. EXPERIMENTATION

A. Human posture database building

In order to evaluate the performance of our posture recognition system, we have to test our system with a joint database for postures. Since, to the best of our knowledge, no joint database for human posture available. Therefore, we create a joint database. We have developed a database capture tool in C++ using Microsoft's Skeleton API. The interface of this tool contains 3 sub windows: Input, Output, Skeleton view. The Input window displays color images captured by RGB camera while Output one displays only a part of image that contains detected human from Input. The Skeleton view window shows the skeleton of detected human using infrared cameras.

With the color information and depth information, we store them in video with the following naming convention:

Color_[Posture]_[Volunteer's_name]_[Recording_time].avi
Depth_[Posture]_[Volunteer's_name]_[Recording_time].avi

For skeleton information, we store it in a text file with naming convention as follows:

[Posture]_[Volunteer's_name]_[Recording_time].txt

Each line of the text file represents joint information as explained above for each frame.

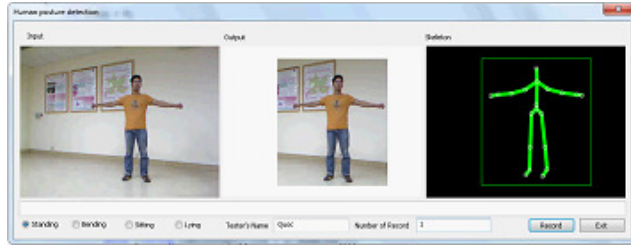


Fig. 7. Interface of the data acquisition program

For this database, the Kinect is set on 110cm measure height. The angle of Kinect for 3 postures “Standing”, “Sitting” and “Bending” is 0 degree and for posture “Lying” is -10 degree. The testing room uses neon light (but actually doesn't effect on the result of recording skeleton data because Kinect use IR camera which works very well without light). The interface is shown in Fig. 7.

We capture four main postures (standing, sitting, bending, and lying) of 5 people (3 men and 2 women). For each person, we record 3 times, each time the duration is about 5 to 8 seconds. Subject is asked to do the posture with the following scenarios:

- *Scenario for standing posture:* subject stands straightly with his/her arms relax. His/her eyes look at Kinect (see Fig. 8 (a)). In the first and the second recording times, the distance between subject and Kinect is 320cm and in the third recording times, the distance is 270 cm.

- *Scenario for bending posture:* subject bends perpendicularly to Kinect, his/her hands are on his/her knees. His/her eyes look at the floor. The first recording times, his/her legs are close (see Fig. 8 (b)). The second and the third recording time, his/her legs open as wide as the shoulders. All three times, the distance between subject and Kinect is 320cm.
- *Scenario for lying posture:* human body is in horizontal direction from the viewpoint of Kinect. His/her legs are straightened. His/her hands are stretched along the body (see Fig. 8 (c)). His/her eyes look at the ceiling. For all three recording times, the distance between subject and Kinect is 320 cm.
- *Scenario for sitting posture:* subject sits on a chair. He/she leans his/her back on the chair and keeps his/her back straight. His/her knees are fold perpendicularly. His/her feet touch the floor. His/her hands are on his/her thigh. His/her eyes look at Kinect (see Fig. 8 (d)). The first and the second recording times, the distance between subject and Kinect is 320cm and the third recording times, the distance is 370 cm.

During the recording, subject remains in the same position. Finally, we have 180 files (60 color videos, 60 depth videos, 60 data text files). Fig. 8 gives some images of four postures in the database.

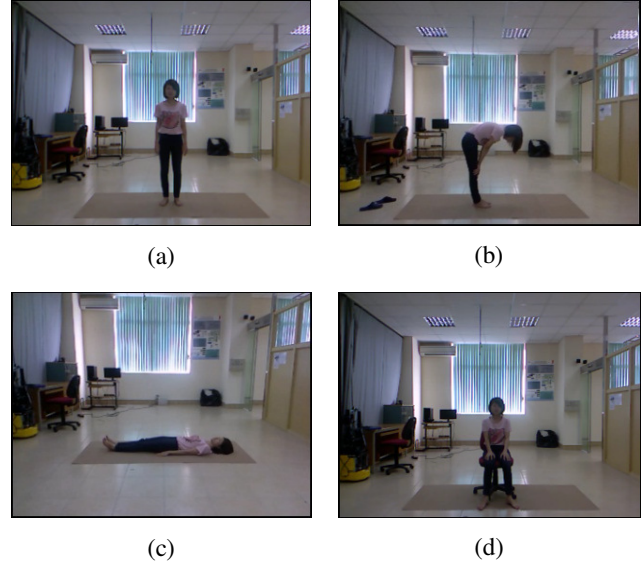


Fig. 8. Four postures (a) standing, (b) bending, (c) lying, (d) sitting

B. Human posture recognition performance analysis

1) Offline evaluation

In this section, we will analyze the recognition of our system in two types of evaluation: offline evaluation and online evaluation.

In our experimentation, we use LibSVM [8]. The type of SVM chosen is nu-SVC and the kernel is RBF (radial basis function). In some experiments, we scale data before applying

SVM, so we can compare the result between scaled data and not-scaled data. For the scaling, the upper value is 1 and the lower value is -1.

Concerning offline evaluation, from the recorded data, we prepare the training data and testing data. There are 1114 postures instances for testing and 1115 postures instances for training.

Experiment 1 (Ex1)- Using absolute coordinate value of joints without scaling: We use the exactly the coordinate of each joints in the skeleton (x, y, z) as feature vector for SVM. Since, the skeleton has 20 joints so we have total 60 features. This experiment gives very high accuracy 100% (1114/1114). But when apply to real time processing, result is not good because depends on the distance of the detected human to the Kinect and the angle of Kinect, the coordinate could be very different from the model.

Experiment 2 (Ex2) - Using 7 joint-angles without scaling: From the 10 important joints defined in Section III.B, we compute 6 angles that are: $(\vec{BA}, \vec{BC})$, $(\vec{CB}, \vec{CD})$, $(\vec{GD}, \vec{GH})$, $(\vec{HG}, \vec{HQ})$, $(\vec{DE}, \vec{DG})$, $(\vec{FE}, \vec{FO})$. Beside these angles, we calculate the angle between vector $\vec{CB}$ and the positive y axis because this angle allows to distinguish “standing” and “lying” postures. With the same number of data training and testing like the experiment 1, the obtained accuracy is 73.43% (818/1114).

Experiment 3 (Ex3) - Using 7 joint-angles with scaling: In this experiment, we employ the same feature as the second experiment. However, we do the data scaling before applying SVM. The average accuracy is increased to 98.3842% (1096/1114). This accuracy is much higher than the accuracy of the second experiment.

Experiment 4 (Ex4) - Using 9 joint-angles without scaling: The 2 above experiments only calculate angles between concatenate bones. In this experiment, we add 2 more angles: $(\vec{GH}, \vec{CB})$, $(\vec{EF}, \vec{CB})$ that represent for the angle between two legs and spine. These angles can make the posture sitting more differ than the other postures. The accuracy of this experiment is 72.711% (810/1114).

Experiment 5 (Ex5) - Using 9 joint-angles with scaling: This experiment uses the same feature as the experiment 4. We do data scaling before applying SVM. In this experiment, the average accuracy is 98.6535% (1099/1114).

Experiment 6 (Ex6) - Using 17 joint angles without scaling: For this experiment, we calculate all possible angles of joints from human skeleton. This experiment takes 9 angles as experiment 4 and 5. We calculate 8 more angles as follows: $(\vec{BC}, \vec{BL})$, $(\vec{LB}, \vec{LM})$, $(\vec{ML}, \vec{MN})$, $(\vec{BC}, \vec{BI})$, $(\vec{IB}, \vec{IJ})$, $(\vec{JI}, \vec{JK})$, $(\vec{HG}, \vec{HQ})$, $(\vec{FE}, \vec{FO})$. The average accuracy is 65.26 (727/1114). This accuracy is lower than the accuracy obtained in the experiments 2 and 4 because of the irrelevant joints using in this experiment.

Experiment 7 (Ex7) - Using 17 joint angles with scaling: By applying data scaling on feature extracted from the experiment 6, we obtain 98.2047% (1094/1114) of accuracy. This result is slightly worse than that of experiments 3 and 5.

The detail results obtained with 7 experiments are presented in Tab. 1. In all experiments, the recognition of bending posture is always higher than that of three other postures. The reason is that the skeleton corresponding to this posture can be seen as a combination of skeletons of three other postures. Therefore, this posture has a high value of accuracy and also a high value of false positives.

Several conclusions and remarks can be extracted from our experiments. Firstly, using data scaling before applying SVM allows to increase the recognition performance. Secondly, in order to recognize 4 postures (sitting, lying, standing, and bending) we can use either 7 joints or 9 relevant joints as feature vector. This feature gives the similar recognition result as all joints positions.

TABLE I. RECOGNITION ACCURACY OF FOUR POSTURES FOR OFFLINE EVALUATION (%)

Accuracy Posture	Ex1	Ex2	Ex3	Ex4	Ex5	Ex6	Ex7
Standing	100	89.67	100	88.93	100	81.18	100
Sitting	100	88.98	100	86.53	100	61.22	100
Lying	100	0	95.43	0	95.43	0	92.53
Bending	100	100	98.04	100	98.88	100	99.43
Average	100	73.43	98.38	72.71	98.65	65.26	98.20

2) Online evaluation

Concerning online evaluation, we create a program that takes human posture instances captured directly from Kinect and predicts label for each instance. The interface of this program is shown in Fig. 9. We ask a subject who does not participate into database building. For 3 postures “Standing”, “Sitting”, “Bending”, we ask him to do these postures in different positions and distances. The test positions are detailed in Tab. 2. The obtained results are presented in Tab. 3. As we can see, the absolute coordinate that gives a very good result in offline evaluation has low recognition accuracy in online evaluation if the position of the subject and the configuration between the subject and the Kinect is different from that in training database. In these cases, the angles are more relevant for posture representation.

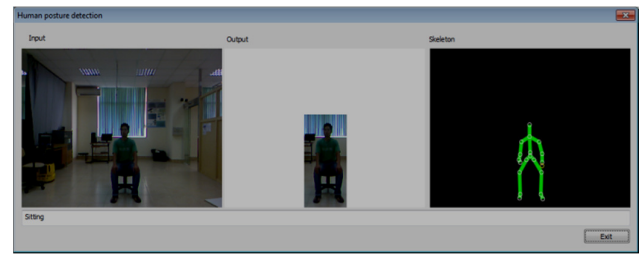


Fig. 9: Interface of the program for online evaluation

TABLE II. TEST POSITIONS

Position	Distance angles	Distance from Kinect	Angle with Kinect
Pos1		320	0
Pos2		320	90 (left)
Pos3		320	90 (right)
Pos4		270	0
Pos5		370	0

TABLE III. RECOGNITION ACCURACY OF FOUR POSTURES FOR ONLINE EVALUATION (%)

Posture	Features	Pos 1	Pos 2	Pos 3	Pos 4	Pos 5
Standing	Coordinate	100	100	100	100	100
	7 angles	100	6.45	100	100	100
	9 angles	100	28.95	51.95	100	100
	17 angles	100	93.06	100	100	100
Sitting	Coordinate	100	22.09	0	100	100
	7 angles	100	100	100	100	100
	9 angles	100	100	86.96	100	100
	17 angles	100	100	100	100	100
Bending	Coordinate	100	0	0	100	100
	7 angles	100	15.79	0	7.22	100
	9 angles	100	53.42	0	98.70	100
	17 angles	100	0	0	4.23	12.12
Lying	Coordinate	12.50	0	0	0	N/A
	7 angles	100	100	75.27	83.84	N/A
	9 angles	98.82	38.27	6.58	49.33	N/A
	17 angles	100	100	50	92.41	N/A

V. CONCLUSIONS

In this paper, we have proposed a method for human posture recognition using skeleton provided by Kinect device. We have conducted 7 different experiments with 4 different features extracted from the tracked human skeleton. The obtained results show this skeleton allows classifying well four postures. Several remarks and recommendations are extracted

from the experimental results. In the future, we will extend this work by analyzing performance human posture recognition with other features as well as modeling and abnormal event.

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